结力基础班第 4 次作业——梁和刚架

梁和刚架作业及答案

2-1 图 2-60 所示结构截面 A 的弯矩（以下侧受拉为正）是（　　）。（西南交通大学 2008）

A. M　　　　B. $-M$

C. $-2M$　　　　D. 0

2-2 图 2-61 所示结构剪力 $F_{QBA}=$________。（重庆大学 2010）

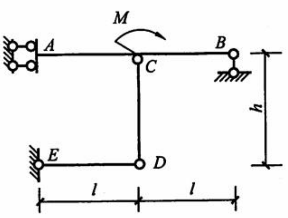


图 2-60

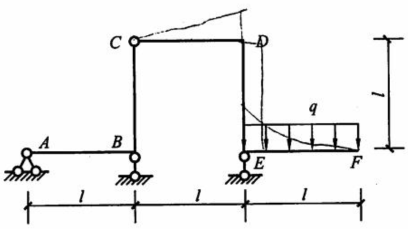


图 2-61

2-3 图 2-62 所示刚架中 AB 杆的弯矩为________，剪力为______。（河北工业大学 2010）

2-4 图 2-63 所示刚架，AD 杆截面 D 弯矩 $M_{DA}=$________（内侧受拉为正）。（浙江大学 2011）

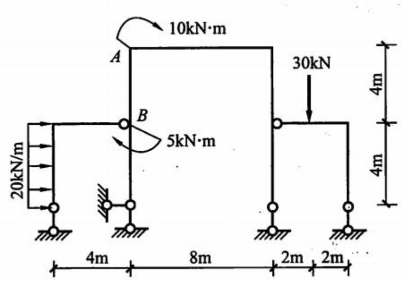


图 2-62

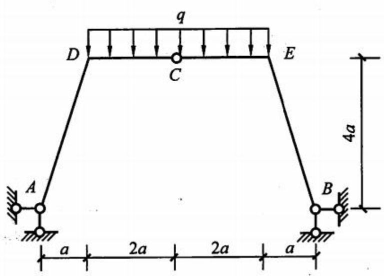


图 2-63

2-5 试作图 2-64 所示结构的弯矩图。（北京科技大学 2008）

2-6 作图 2-65 所示多跨梁的弯矩图和剪力图，并给出 AB 杆的受力平衡图。（北京交通大学 2011）

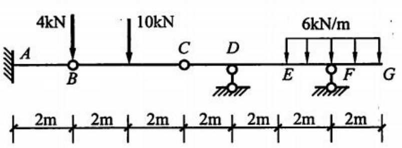


图 2-64

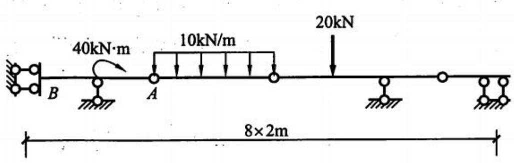


图 2-65

2-7 求图 2-66 所示结构的内力、作弯矩图（可不写过程）。（东南大学 2008）

2-8 绘出图 2-67 所示结构的弯矩图。（武汉大学 2010）

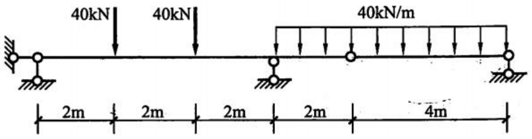


图 2-66

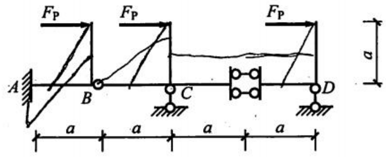


图 2-67

2-9 作图 2-68 所示静定刚架 M 图。（福州大学 2008）

2-10 绘制图 2-69 所示结构的弯矩图。（武汉科技大学 2009）

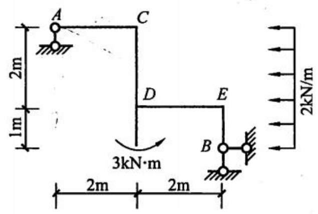


图 2-68

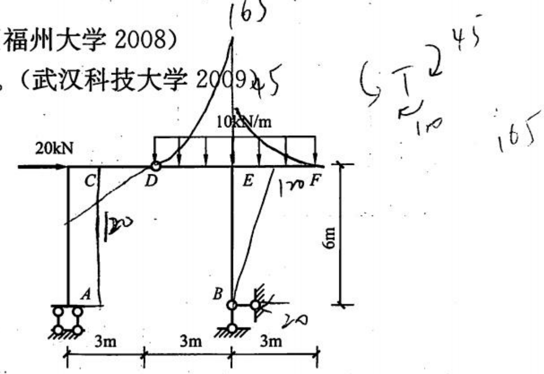


图 2-69

2-11 绘图 2-70 所示结构的弯矩图。（重庆大学 2012）

2-12 已知图 2-71 所示结构的弯矩图，试作剪力图和轴力图。已知 $F_P=10\text{kN}$，$M=40\text{kN}\cdot\text{m}$。（哈尔滨工业大学 2010）

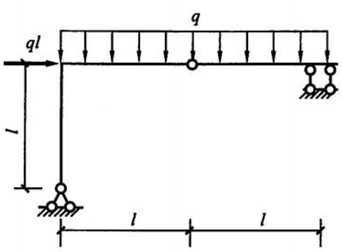


图 2-70

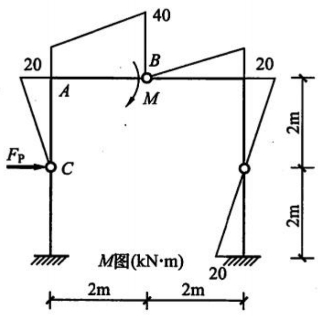


图 2-71

2-13 作图 2-72 所示刚架的弯矩图。（中南大学 2009）

2-14 绘出图 2-73 所示结构的弯矩图。（武汉大学 2010）

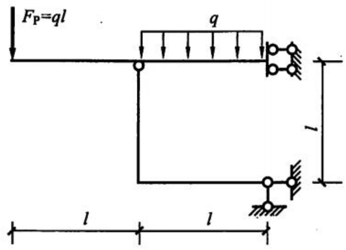


图 2-72

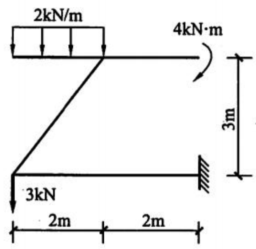


图 2-73

2-15 作图 2-74 所示结构的内力图。(北京工业大学 2011)

2-16 求图 2-75 所示结构的内力,并作弯矩图。(可不写过程)(东南大学 2008)

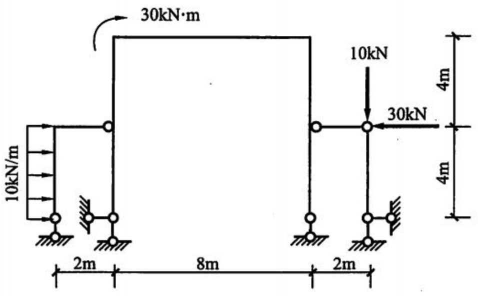


图 2-74

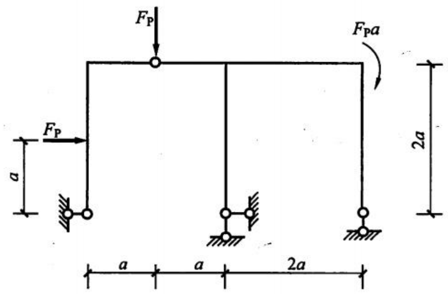


图 2-75

2-17 作图 2-76 所示结构的弯矩图。(无须写出分析过程)(东南大学 2007)

2-18 作图 2-77 所示结构弯矩图。(中南大学 2010)

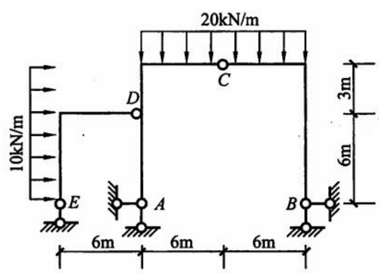


图 2-76

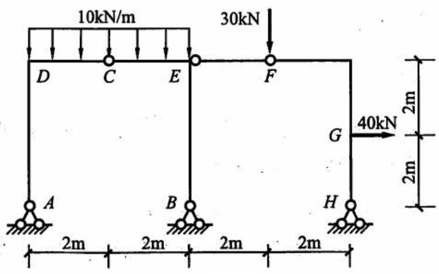


图 2-77

2-19 已知结构在集中力和集中力偶矩作用下的弯矩图如图 2-78 所示,其中 BC 段为二次抛物线,试确定结构所受的荷载情况,并绘出 BE 段隔离体受力平衡图。(北京交通大学 2009)

2-20 已知图 2-79 所示静定刚架的弯矩图(曲线部分为二次抛物线),试绘图标出作用于刚架上的荷载,并作出剪力图和轴力图(假设各杆无局部平衡的轴向荷载,无分布力偶)。(浙江大学 2012)

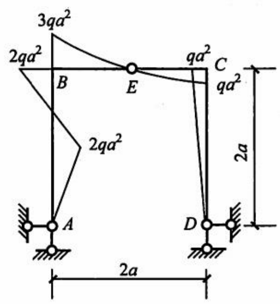


图 2-78

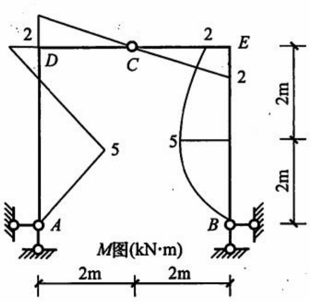


图 2-79

2－21 不经过计算，绘制图 2－80 所示结构弯矩图。(武汉科技大学 2009)

2－22 试求解图 2－81 所示刚架，并绘制弯矩图。(东北大学 2005)

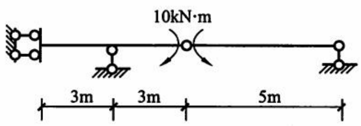


图 2－80

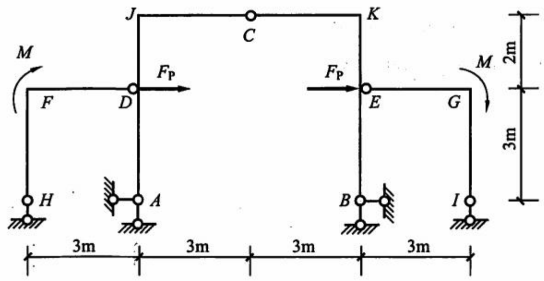
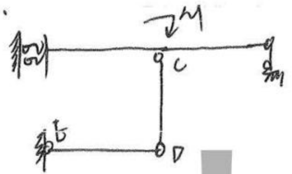


图 2－81

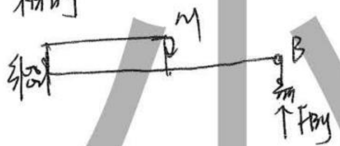
2-1.



$M_C=0$ ∴ $N_{ED}=0$

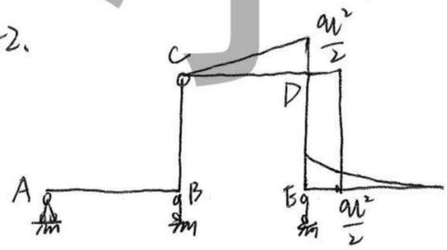
∴ $N_{CD}=0$

相当于



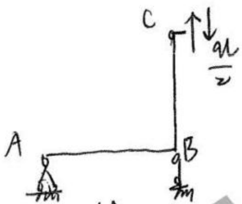
由 $\Sigma Y=0$ ∴ $F_{By}=0$

2-2.



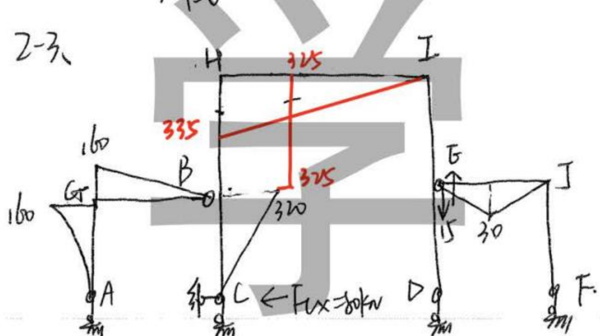
$M_E = ql \times \frac{l}{2} = \frac{ql^2}{2}$

DE杆无剪力



∴ $Q_{BA}=0$.

2-3.



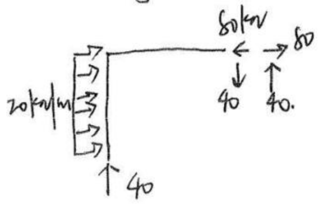
$\Sigma X=0$, $F_{XC}=20\times4=80kN$

$M_{GA}=20\times4\times2=160kN\cdot m$

$M_F=M_J=0$. $\frac{1}{4}\times30\times4=30kN\cdot m$

$M_B=0$

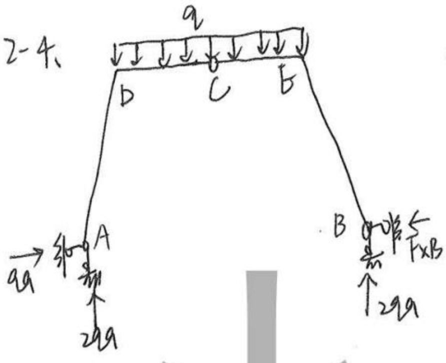
$F_{Ay}\times4=20\times4\times2$ ∴ $F_{Ay}=40kN$



$M_H=80\times8-80\times4=320\ kN\cdot m$

$M_I=0$

2-4.



$M_A=0$.

$F_{yB}\times 6a = q\times 4a\times 3a$

$F_{yB} = 2qa$

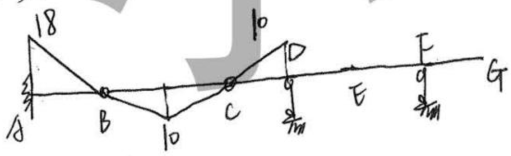
$M_C=0$

$F_{xB}\times 4a + q\times 2a\times a = 2qa\times 3a$

$\therefore F_{xB} = qa$.

$\therefore M_D = qa\times 4a - 2qa\times a = 2qa^2$（外侧受拉）

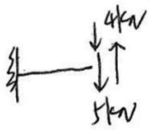
2-5.



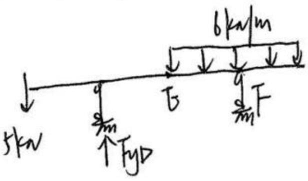
BC为附属部分

AB与CDFG为基本部分

$\frac{1}{4}\times 10\times 4 = 10\text{kN·m}$



$\therefore M_A = 9\times 2 = 18\ \text{kN·m}$

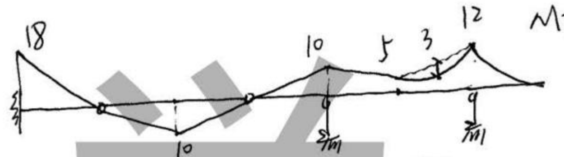


$\Sigma M_F=0$

$F_{yD}\times 4 = 5\times 6$

$F_{yD} = \frac{30}{4} = \frac{15}{2}\text{kN}$

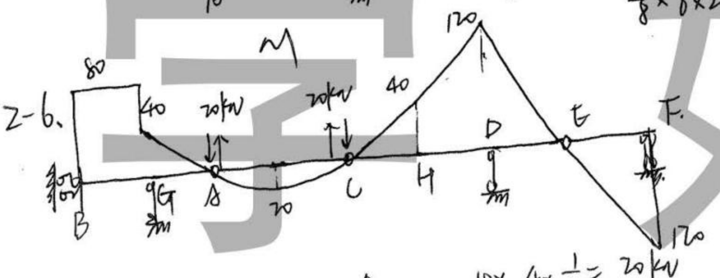
$M_E = \frac{15}{2}\times 2 - 5\times 4 = -5\text{kN·m}$.



$M_F = 6\times 2\times 1 = 12\ \text{kN·m}$

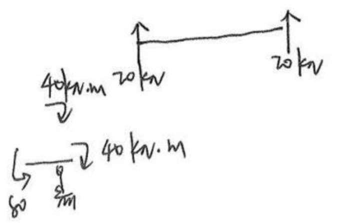
$\frac{1}{8}\times 6\times 2^2 = 3$

2-6.



AC为附属部分

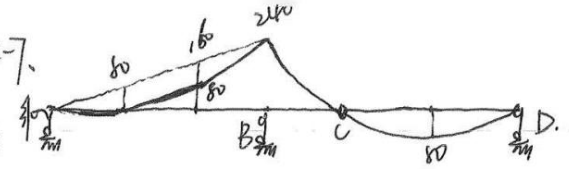
$10\times 4\times \frac{1}{2} = 20\text{kN}$



$\frac{1}{8}\times 10\times 4^2 = 20\text{kN·m}$

$M_D = 20\times 4 + 20\times 2 = 20\times 6 = 120\ \text{kN·m}$

2-7、

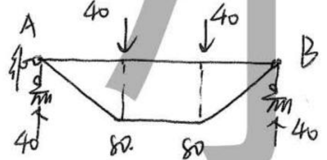


$40\times4\times\frac{1}{2}=80kN$

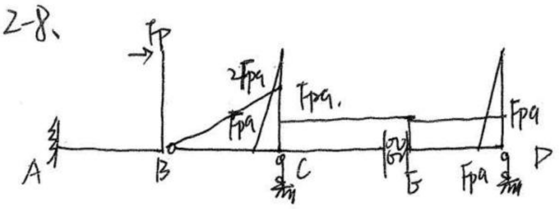
$\frac{1}{8}\times40\times4^2=80kN\cdot m$

AB部分用叠加法
叠加对应简支梁的弯矩

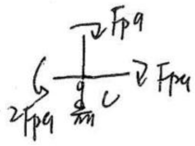
$M_B=80\times2+40\times2\times1=240kN\cdot m$

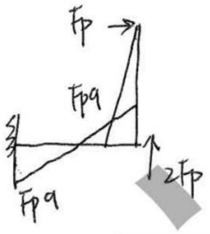


2-8、



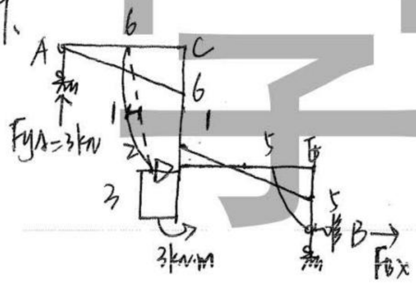
DE杆无剪力，M为常数





$M_A=2F_P\times a-F_P\times a=F_Pa$

2-9、



$M_B=0$

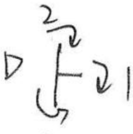
$F_{yA}\times4=3+2\times3\times\frac{3}{2}=12$

$\therefore F_{yA}=3kN$

$M_{DC}=3\times2-2\times2\times1=2kN\cdot m$

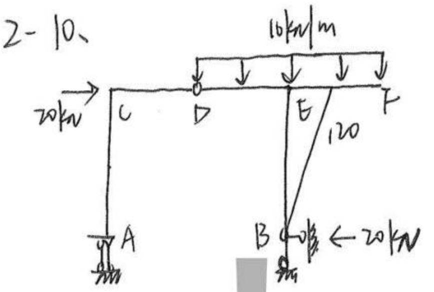
$\frac{1}{8}\times2\times2^2=1$

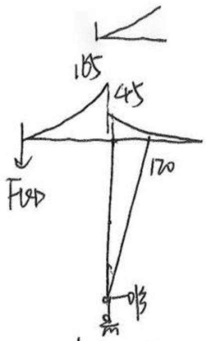
$F_{Bx}=2\times3=6kN$



$\therefore M_E=6\times1-2\times1\times\frac{1}{2}=6-1=5kN\cdot m$

2-10、



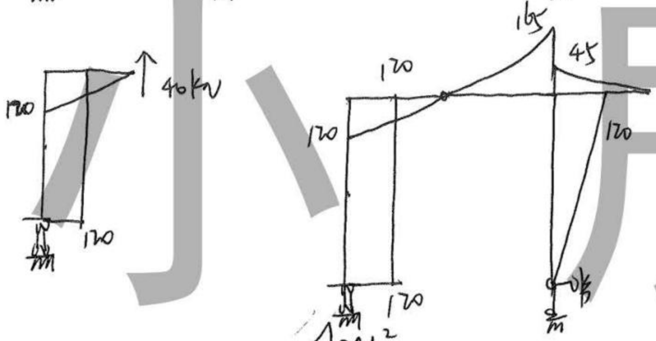


$10\times3\times\frac{3}{2}=45$

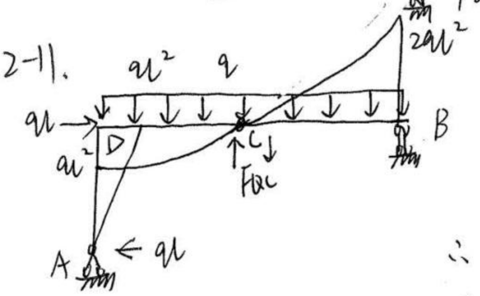
$M_{ED}=165$

$10\times3\times\frac{3}{2}+F_{QD}\times3=165$

$\therefore F_{QD}=40kN$



2-11.

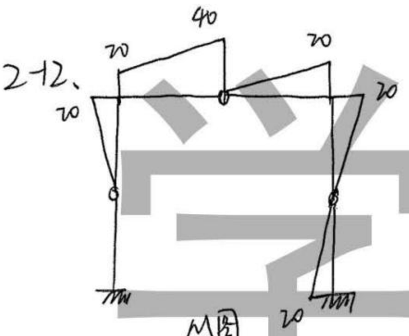


$M_D=ql^2$

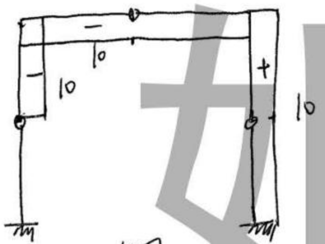
$\therefore F_{QC}\times l-ql\times\frac{l}{2}=ql^2$

$\therefore F_{QC}=\frac{3ql}{2}$ $\therefore M_B=\frac{3ql}{2}\times l+ql\times\frac{l}{2}=2ql^2$

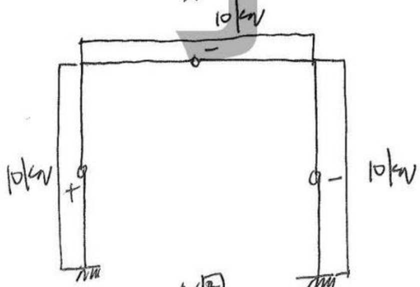
2-12、



M图



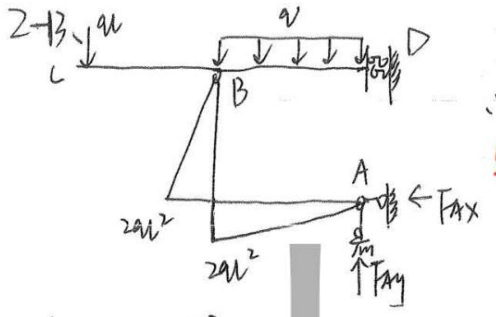
Q图



N图

$\Sigma Y=0$

2-13、

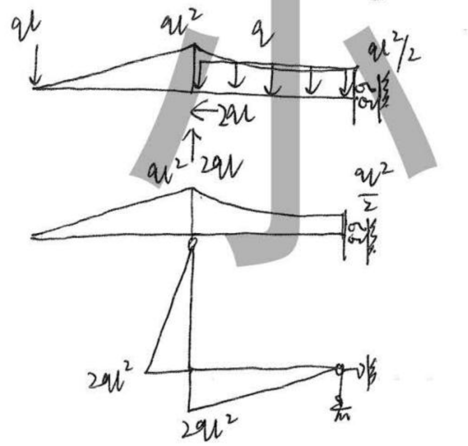


$\Sigma Y=0$

$\therefore F_{Ay}=2ql$

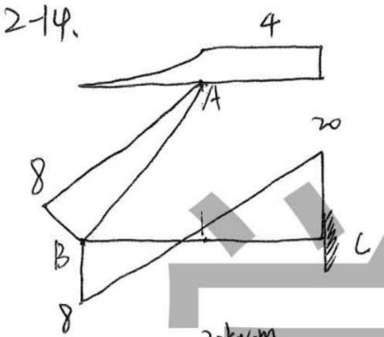
$\Sigma M_B=0$

$F_{Ax}=F_{Ay}=2ql$



$M_D=ql\times 2l+ql\times\frac{l}{2}-2ql\times l=\frac{ql^2}{2}$

2-14、

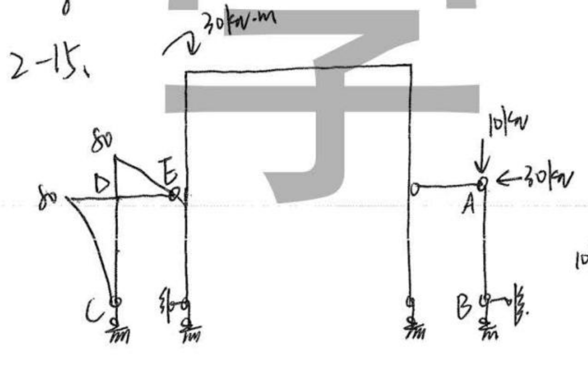


$2\times2\times1=4$

$M_B=4+2\times2\times1=8$

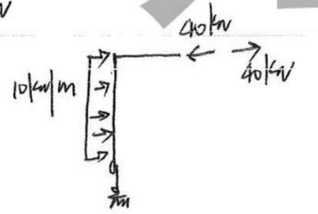
$M_C=3\times4+2\times2\times3-4=20.$

2-15、

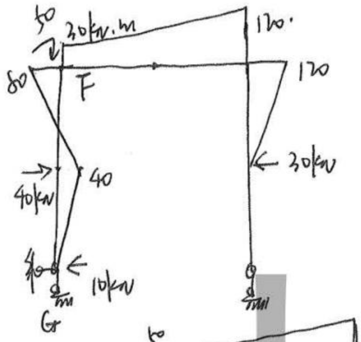


$M_A=0 \therefore F_{Bx}=0.$

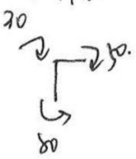
$10\times4\times2=80$ kN·m.

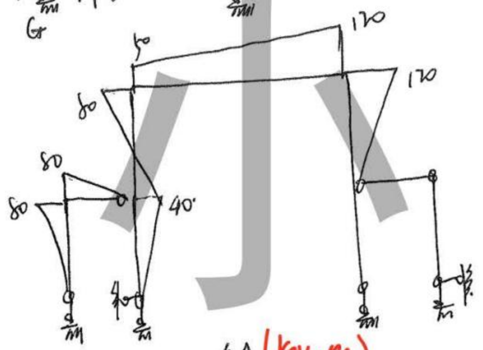


40×4−10×8= 10×10×8=

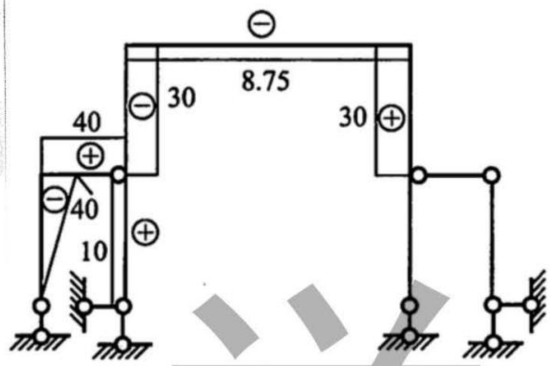


M_{FG}= 40×4−10×8= 80 kN·m.

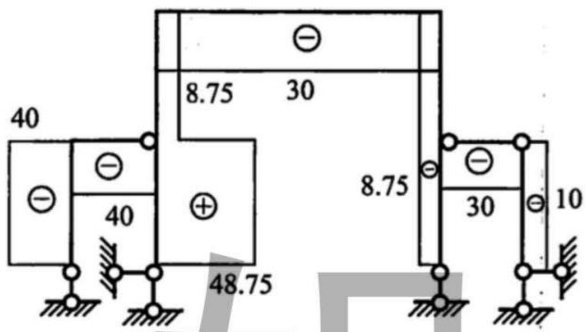




M(kN·m)

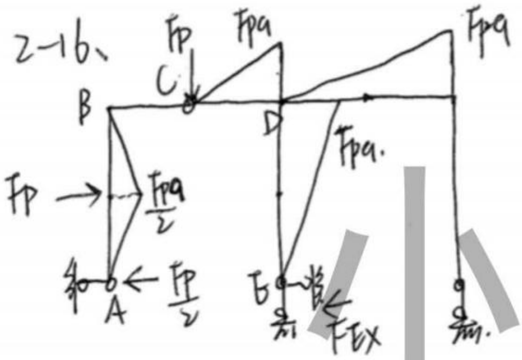


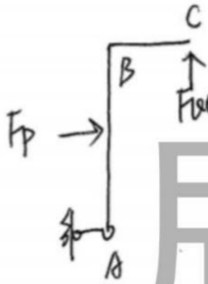
F_Q图(kN)



F_N图(kN)

2-16、





由$\Sigma Y=0$

可知$F_{bc}=0$

$\therefore M_C=M_B=0$

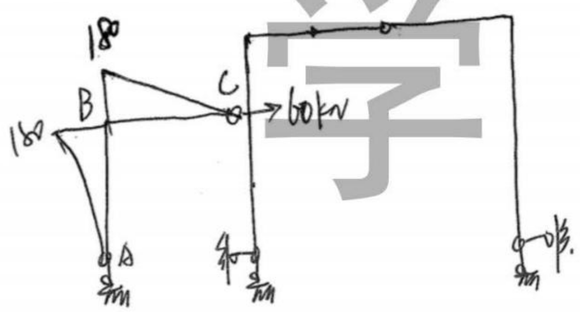
$\frac{1}{4}\times F_p\times 2a=\frac{F_pa}{2}$

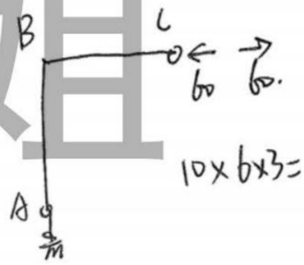
$\therefore M_{DC}=F_p\times a=F_pa.$

$\therefore$ 由$\Sigma X=0$ $\therefore F_{EX}=\frac{F_p}{2}$

二-1

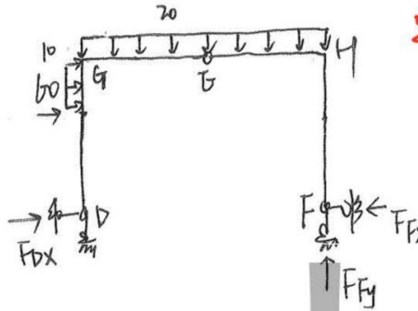
2-17、





$10\times6\times3=180$ 180 180

$\Sigma M_D=0$　　$60\times7.5+20\times12\times6$



$\Sigma M_D=0$

$F_{Fy}\times12=60\times6+10\times3\times7.5+20\times12\times6$

$\therefore F_{Fy}=168.75kN$

取EHF段 $\Sigma M_E=0$

$F_{Fx}\times9+20\times6\times3=168.75\times6$

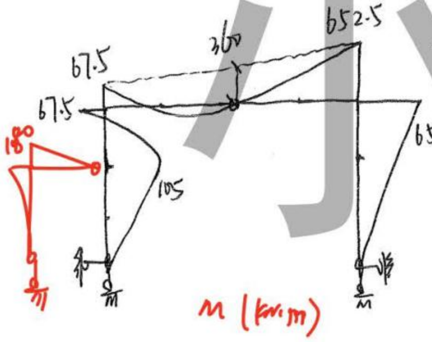
$\therefore F_{Fx}=72.5kN$

$\Sigma X=0$　$F_{Dx}+60+10\times3=72.5$

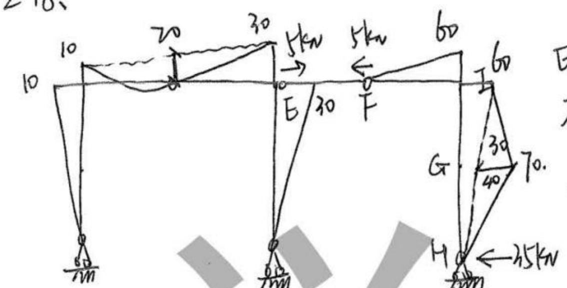
$\therefore F_{Dx}=-17.5kN$

$M_G=60\times3+10\times3\times\frac{3}{2}-17.5\times9=67.5$

$\frac{1}{8}\times20\times12^2=360$



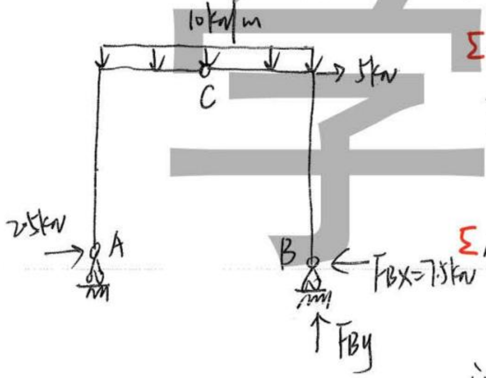
2-18、



EF杆为二力杆
不绘FI传递剪力

$\frac{1}{4}\times40\times4=40$

$\therefore F_{Hx}=35kN$（由M_G可得）



$\Sigma M_A=0$

$F_{By}\times4=10\times4\times2+5\times4$

$F_{By}=25kN$

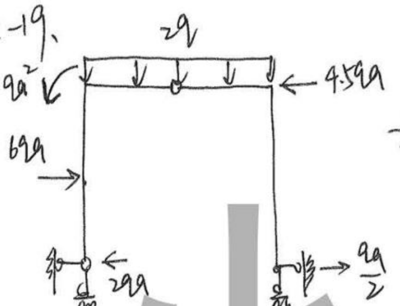
$\Sigma M_C=0$

$F_{Bx}\times4+10\times2\times1=F_{By}\times2=25\times2$

$\therefore F_{Bx}=7.5kN$

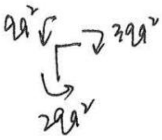
$\frac{1}{8}\times10\times4^2=20.$

2-19、



$\frac{1}{4}\times F\times 2a = qa^2 + 2qa^2 = 3qa^2$

$\therefore F = 6qa$



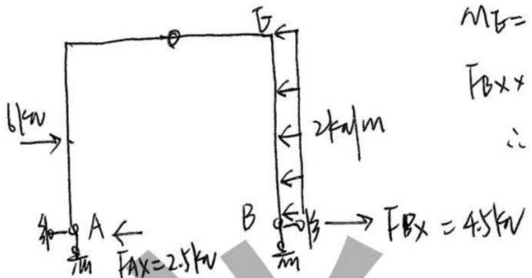
$\frac{1}{8}\times q_1\times(2a)^2 = \frac{1}{2}(3qa^2 - qa^2)$

$\therefore q_1 = 2q$

2-20、

$\frac{1}{4}\times F\times 4 = 1+5 \quad \therefore F = 6kN$

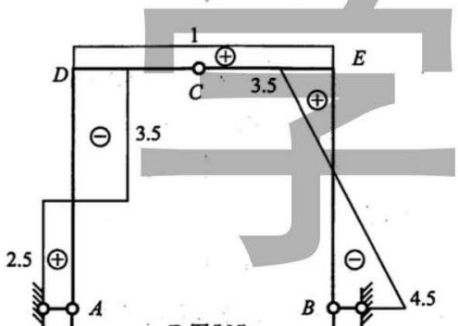
$\frac{1}{8}\times q\times 4^2 = 5-1 \quad \therefore q = 2kN/m$



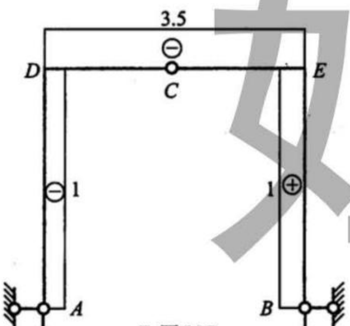
$M_E = 2kN\cdot m$

$F_{Bx}\times 4 - 2\times 4\times 2 = 2$

$\therefore F_{Bx} = 4.5kN$

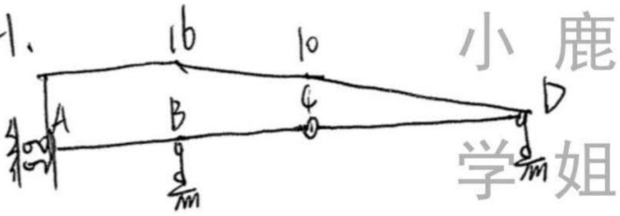


F_Q图(kN)



F_N图(kN)

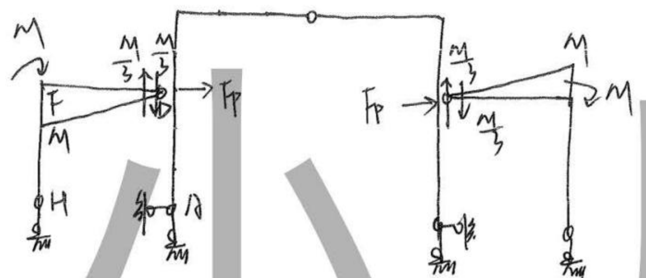
2-21.



BC与CD杆弯矩斜率不变

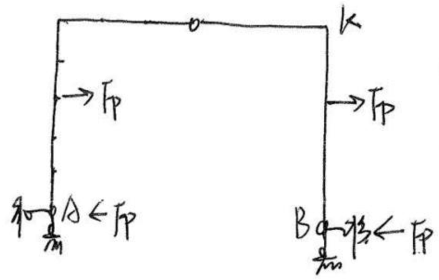
AB杆无剪力，M为常数

2-22.



FH杆M=0，HFD部分不向中间部分传递水平力

而沿杆轴方向作用的力不引起弯矩



由于荷载反对称，

易知A、B处水平反力

$M_K = F_P\times5 - F_P\times2 = 3F_P$.